



SDM COLLEGE OF ENGINEERING AND TECHNOLOGY, Dharwad-580002

(An autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi – 590018)

Project entitled by
“COMMON GATE AMPLIFIER”

TEAM PROJECT BY

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ACKNOWLEDGEMENT

It is our privilege to express our sincerest regards to our Alumni Mr. Praveen Purohit ,Mr. Vishwanath and also Mr. Sharath (currently working in Tessolve Semiconductor PVT LTD), for their guidance, encouragement and whole-hearted cooperation throughout our project.

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INTRODUCTION

- The Common Gate NMOS Amplifier is a type of amplifier circuit that uses an NMOS transistor as the active device.
- In this configuration, the gate terminal of the NMOS transistor is connected to the input ground, while the source terminal is connected to input signal. The drain terminal is connected to the output load.

MOTIVATION

- Understanding NMOS transistor behavior: To gain hands-on experience with NMOS transistors and understand their characteristics.
- Understanding transient and AC analysis of common gate amplifier.
- Improving circuit design skills: To develop skills in circuit design, simulation, and testing.

OBJECTIVES

- Design and implement a common gate amplifier using Cadence and LT spice Tool.
- Analyze the circuit characteristics and performance.
- Further extending our project to implement BGR(Band Gap Reference) circuit.

LITERATURE SURVEY

1. This article analyses the performance of Metal Oxide Semiconductor Field Effect Transistor (MOSFET) in various electronic systems and power electronics.
2. It discusses the use of asymmetric inner surface oxides, layout optimization of bipolar MOSFETs, design enhancements of fully-encircled gate MOSFETs, and double-layer epitaxial structures.

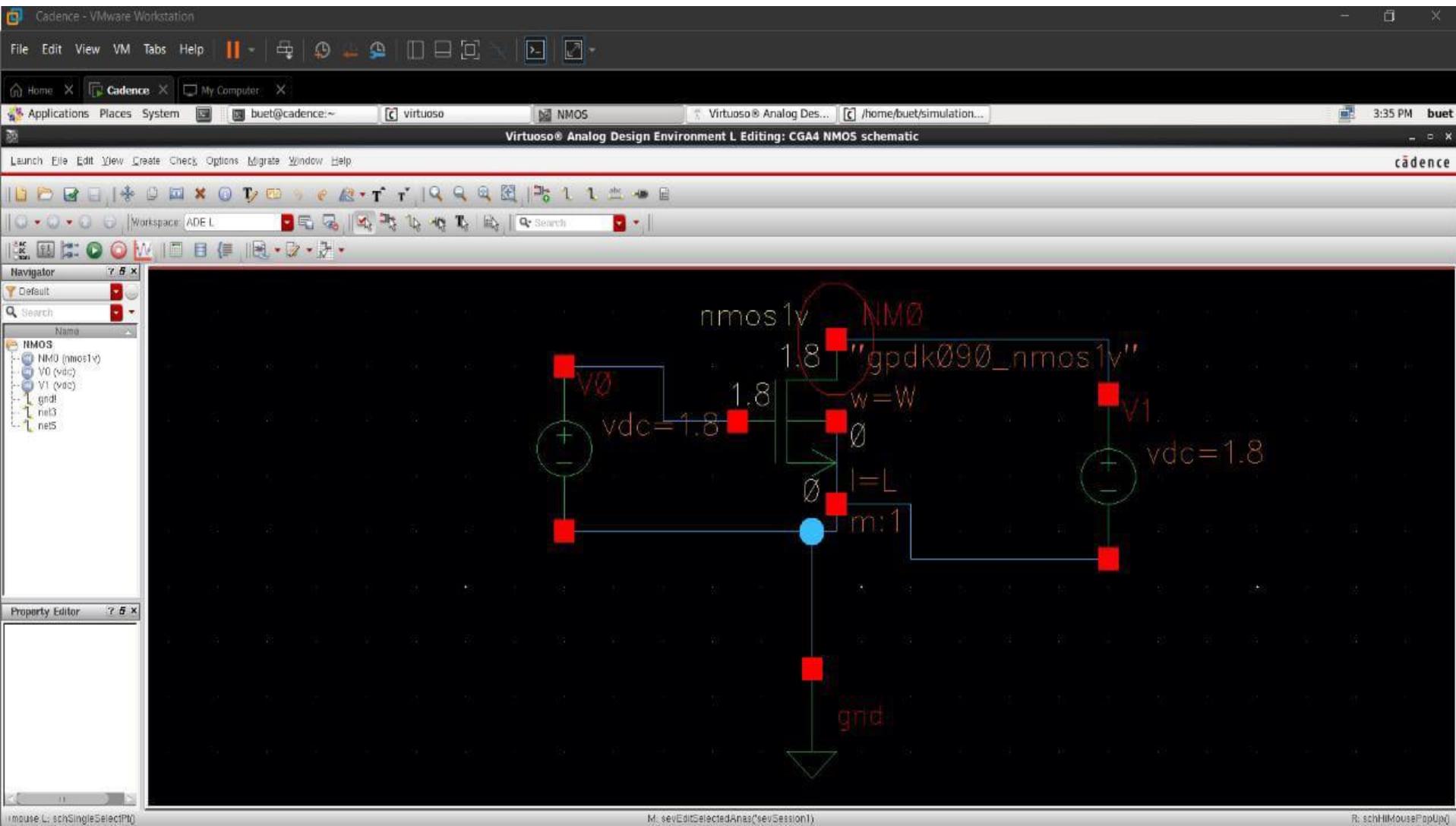
SIGNIFICANCE OF USING NMOS FOR COMMON GATE AMPLIFIER

- High transconductance: NMOS transistors typically exhibit higher transconductance compared to PMOS transistors for the same size and current.
- Fast Operation: NMOS devices generally have higher electron mobility than PMOS devices (since electrons are faster carriers than holes).

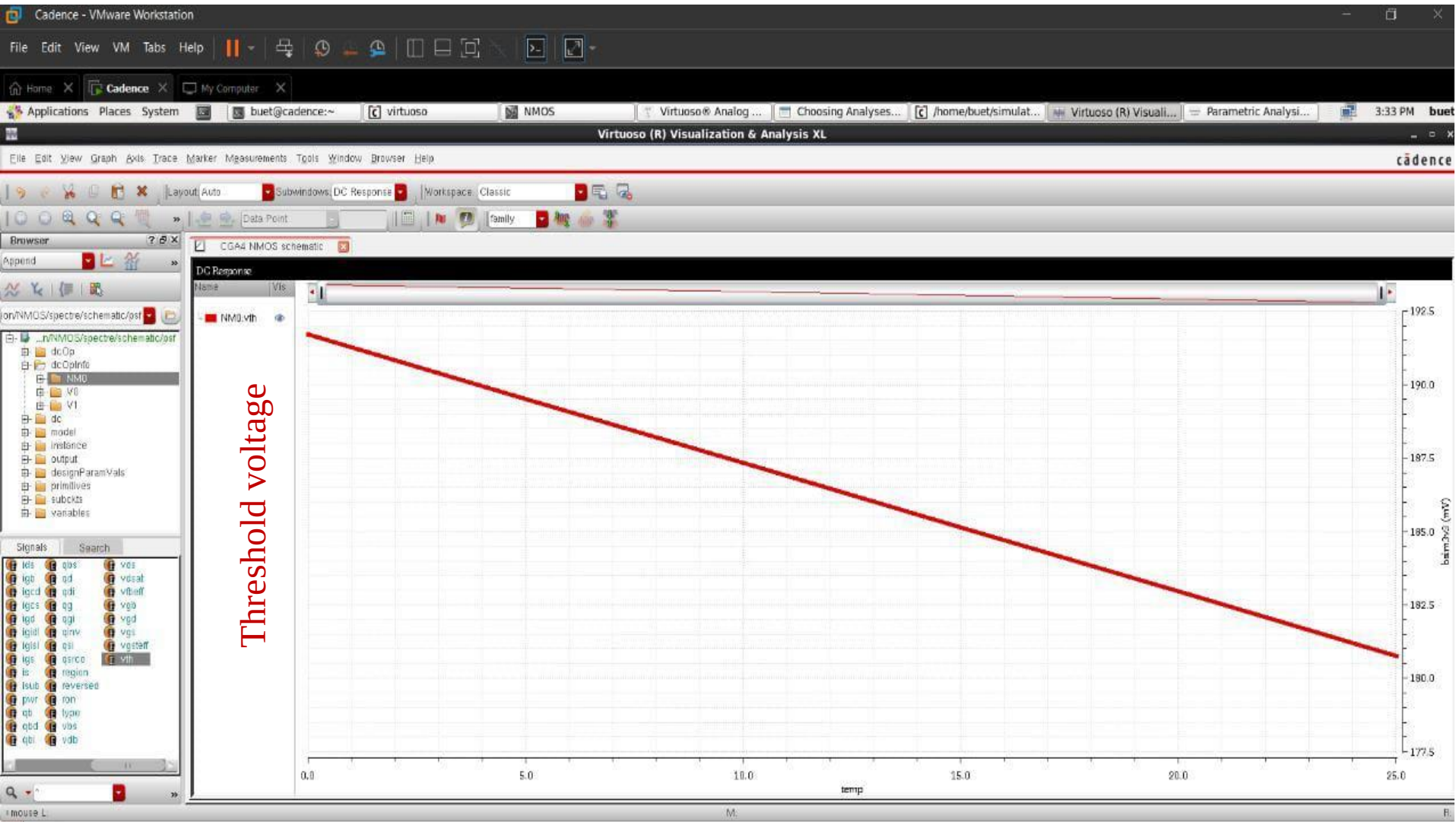
SIGNIFICANCE OF USING NMOS FOR COMMON GATE AMPLIFIER

- Low Input Impedance: In the common-gate configuration, the input impedance is inherently low, which is ideal for applications like wideband RF amplifiers or current buffers.
- Biasing Simplicity: NMOS transistors are easier to bias in the active region (especially for the common-gate configuration) because they require positive voltages relative to the source,

NMOS ANALYSIS



Threshold voltage vs Temperature

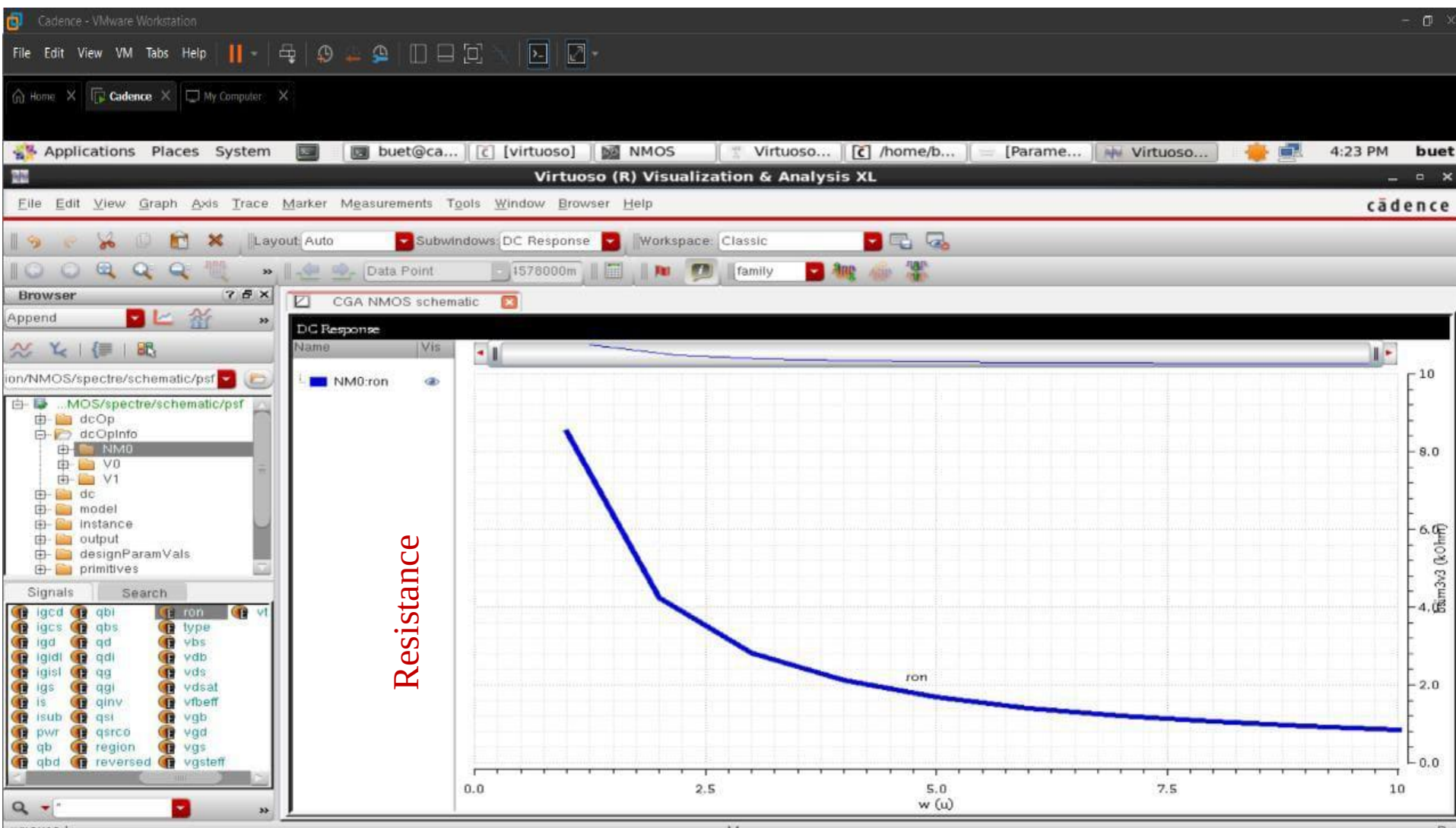


Temperature

Threshold voltage vs Temperature

- As Temperature increases Threshold voltage (V_{th}) decreases.
- Due to increase in the temperature electrons become more energetic and so relative barrier is lower, hence threshold voltage decreases.

Width vs Resistance

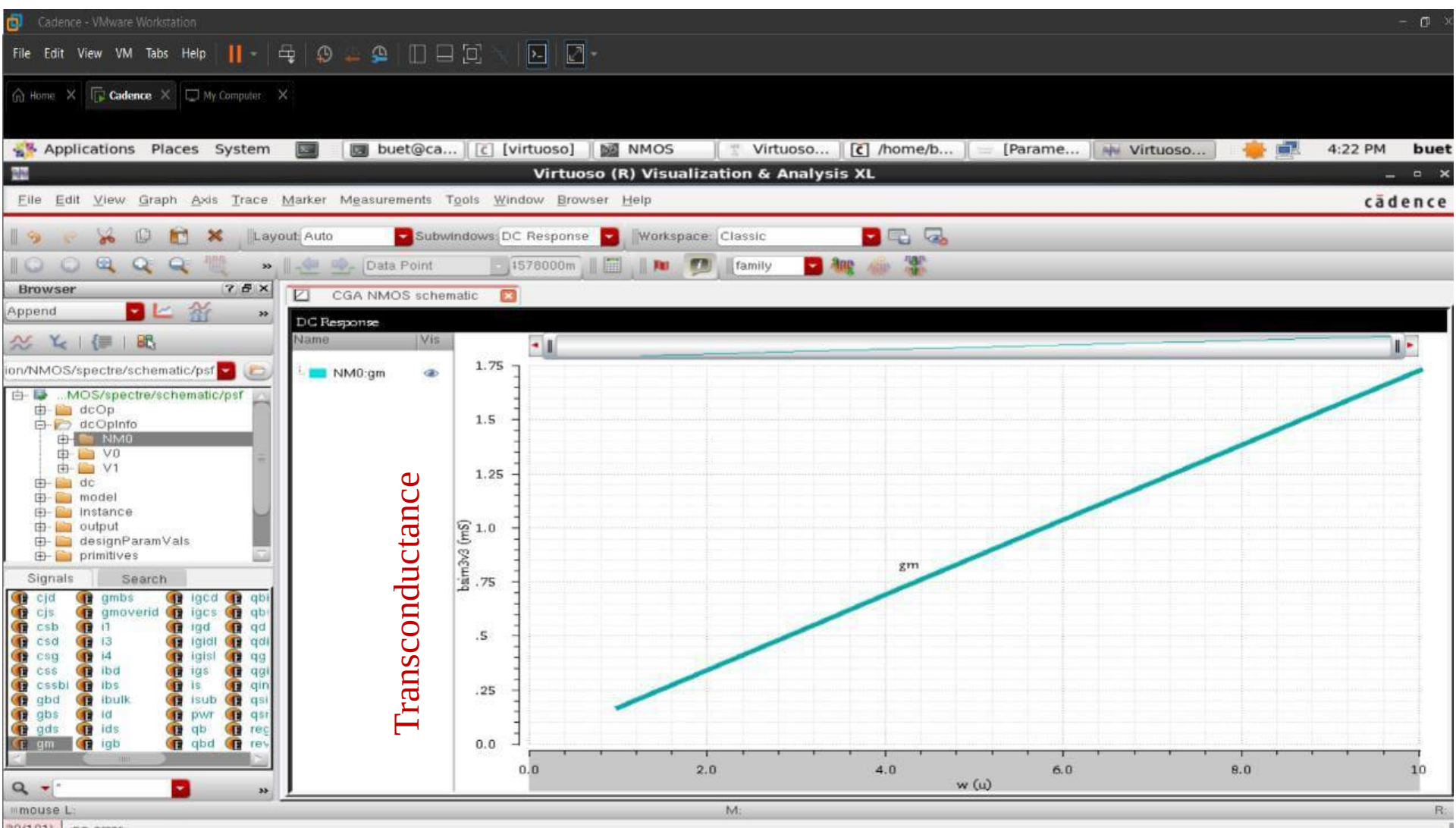


Width

Width vs Resistance

- Width is inversely proportional to resistance.
- As Width increases resistance decreases.

Width vs Transconductance(gm)

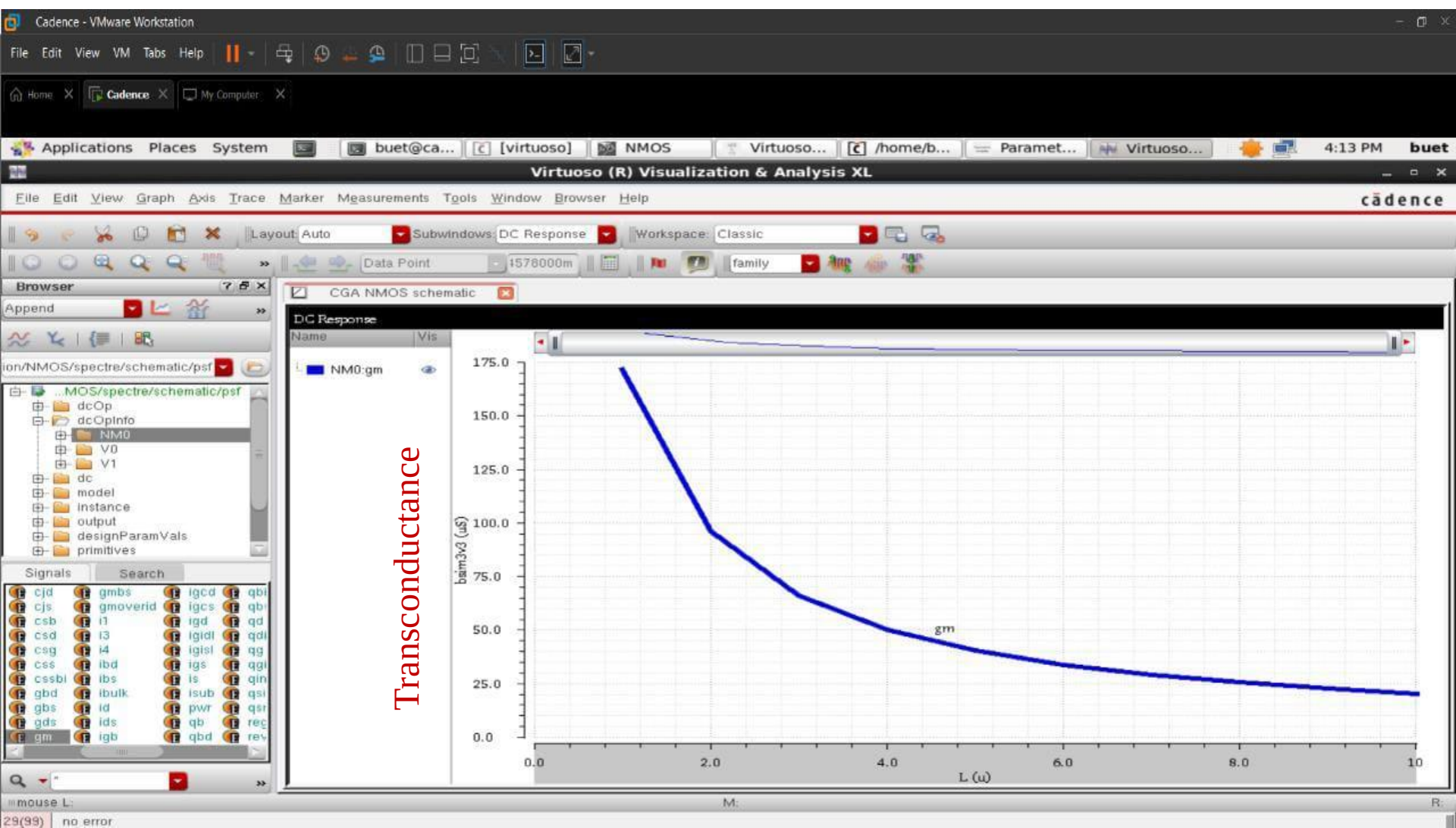


Width

Width vs Transconductance(g_m)

- Transconductance factor is directly proportional to width.
- As Width increases Transconductance factor also increases.
- This is because larger width allows more I_d to flow for a given V_{gs} .
- g_m measures how effectively the transistor can control I_d with V_{gs} .

Length vs Transconductance (gm)

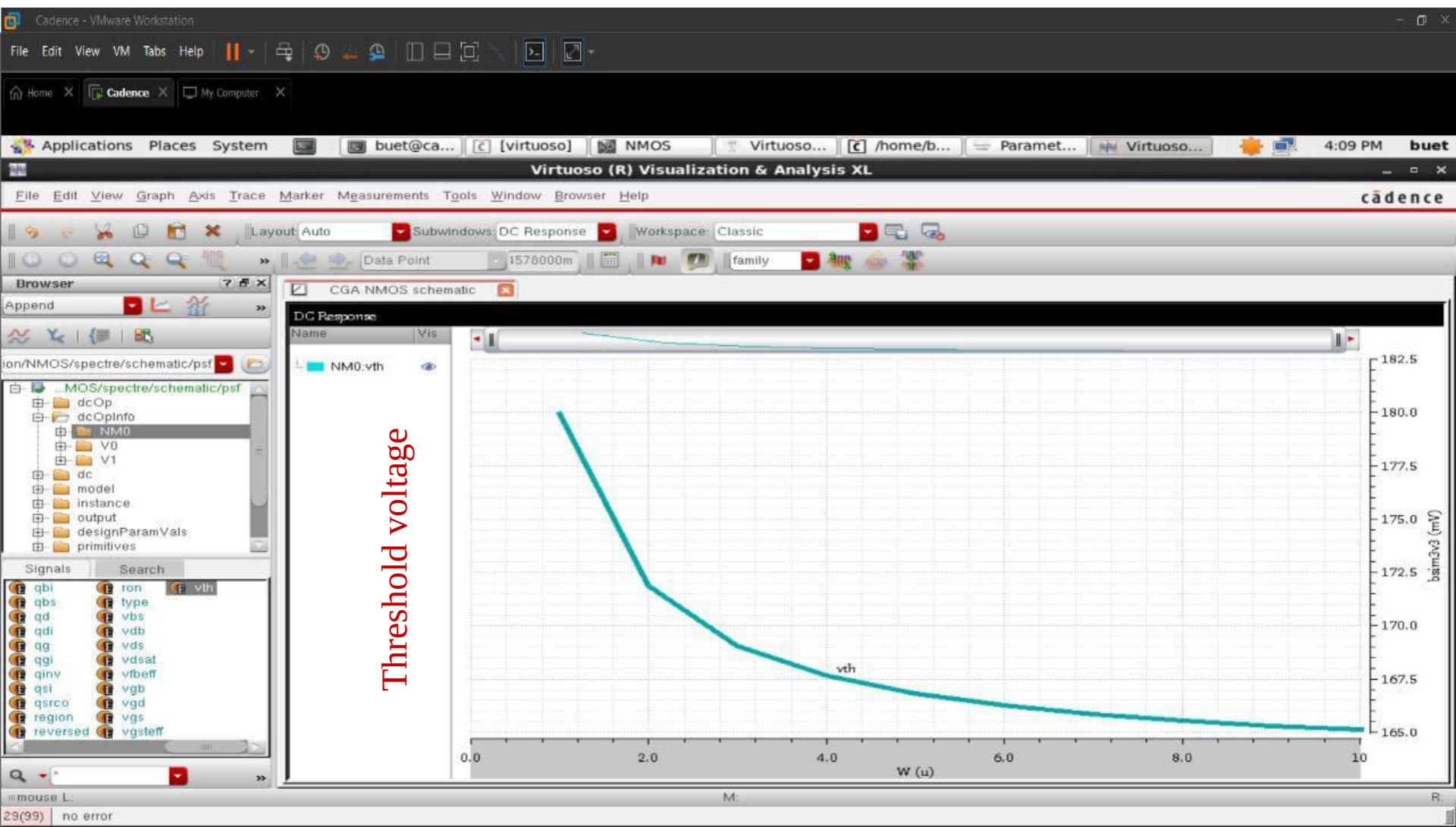


Length

Length vs Transconductance (g_m)

- As Length of the channel between source and drain increases, transconductance decreases.
- This is because shorter length of channel allows a higher electric field between source and drain which in turn increases I_d for given V_{gs} , enhancing g_m .

Width vs Threshold voltage(V_{th})



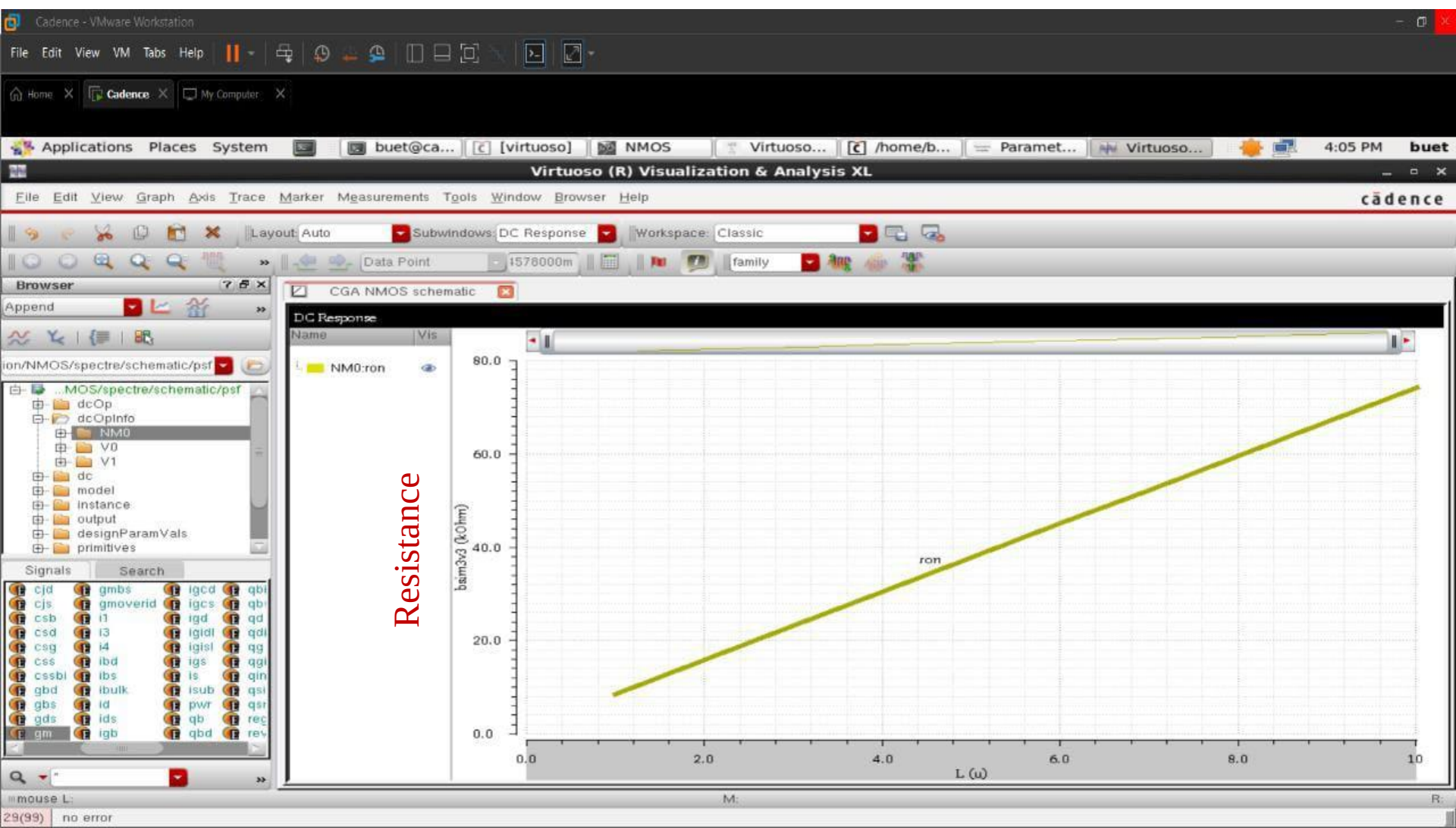
Threshold voltage

Width

Width vs Threshold voltage(V_{th})

- As Width increases Threshold voltage(V_{th}) decreases.
- Due to reduced channel resistance.

Length vs Resistance



Length vs Resistance

- Length is directly proportional to Resistance.
- As Length of the channel between source and drain increases Resistance is also increases.

Width vs Drain current(I_d)

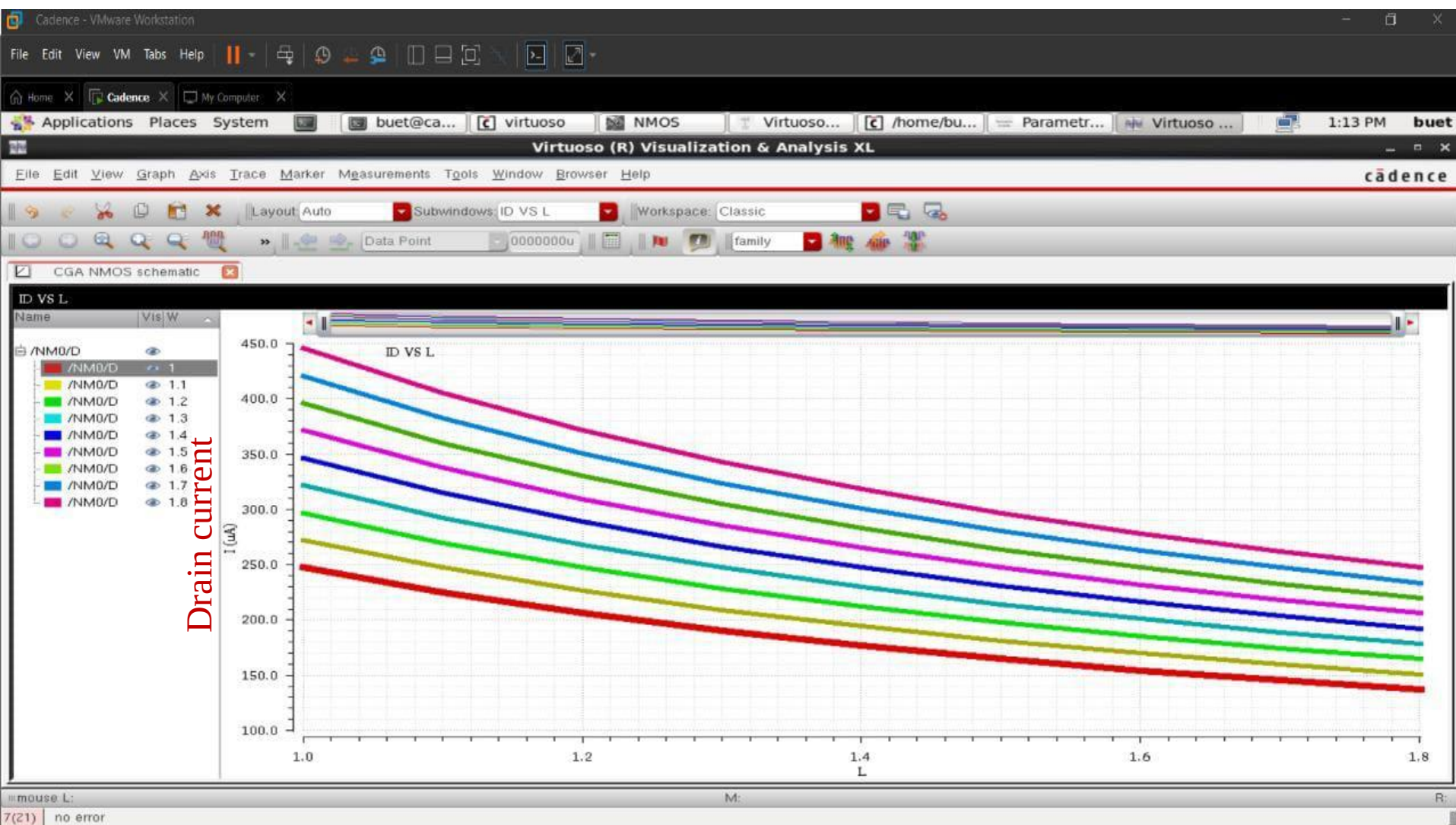


Width

Width vs Drain current(I_d)

- Width is directly proportional to drain current.
- As width increases drain current also increases.

Length vs Drain current

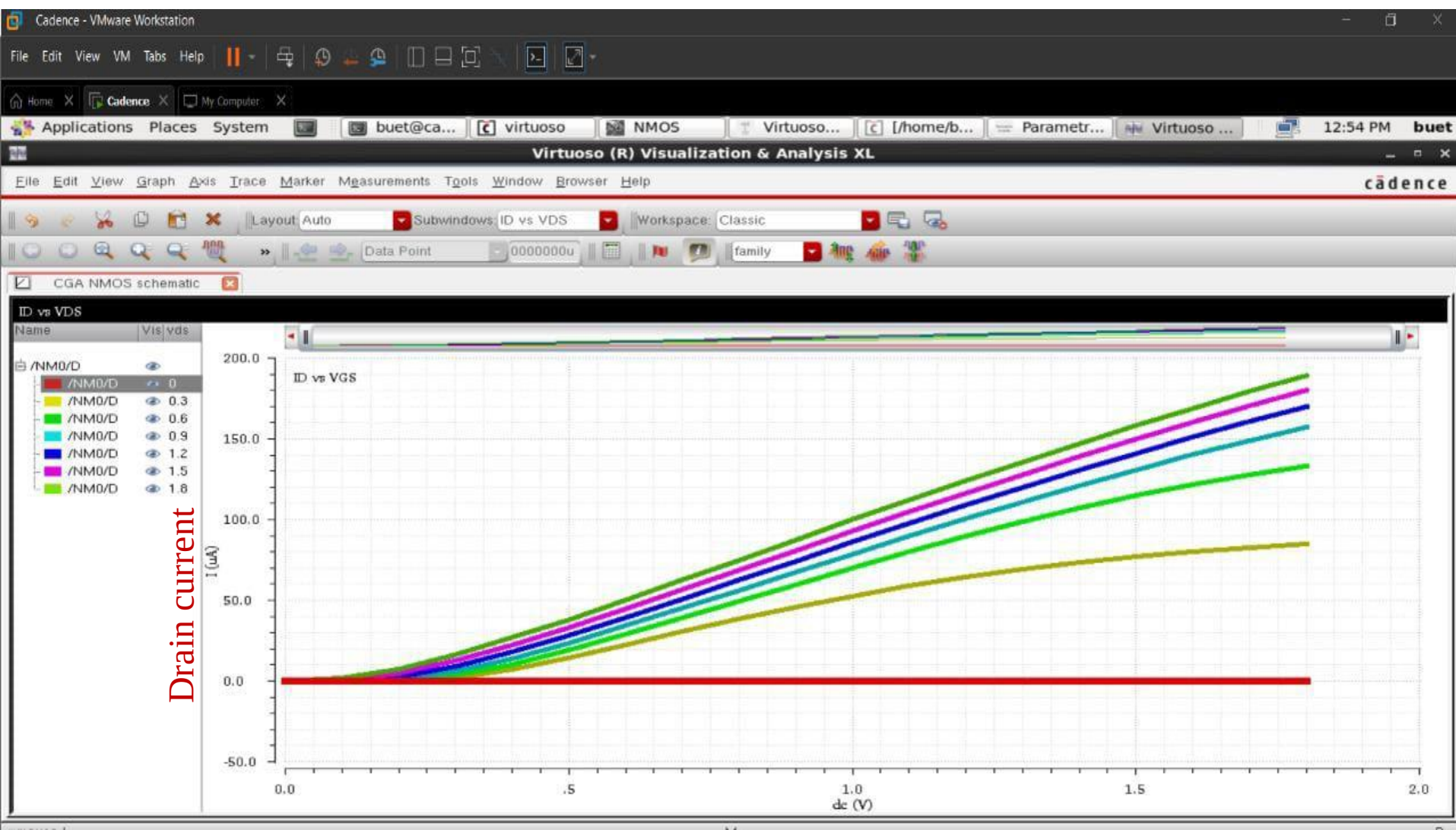


Length

Length vs Drain current

- As Length of the channel increases drain current will decrease.
- This is because resistance is proportional to length, a shorter channel has less resistance and higher current.

TRANSFER CHARACTERISTICS(ID vs VGS)

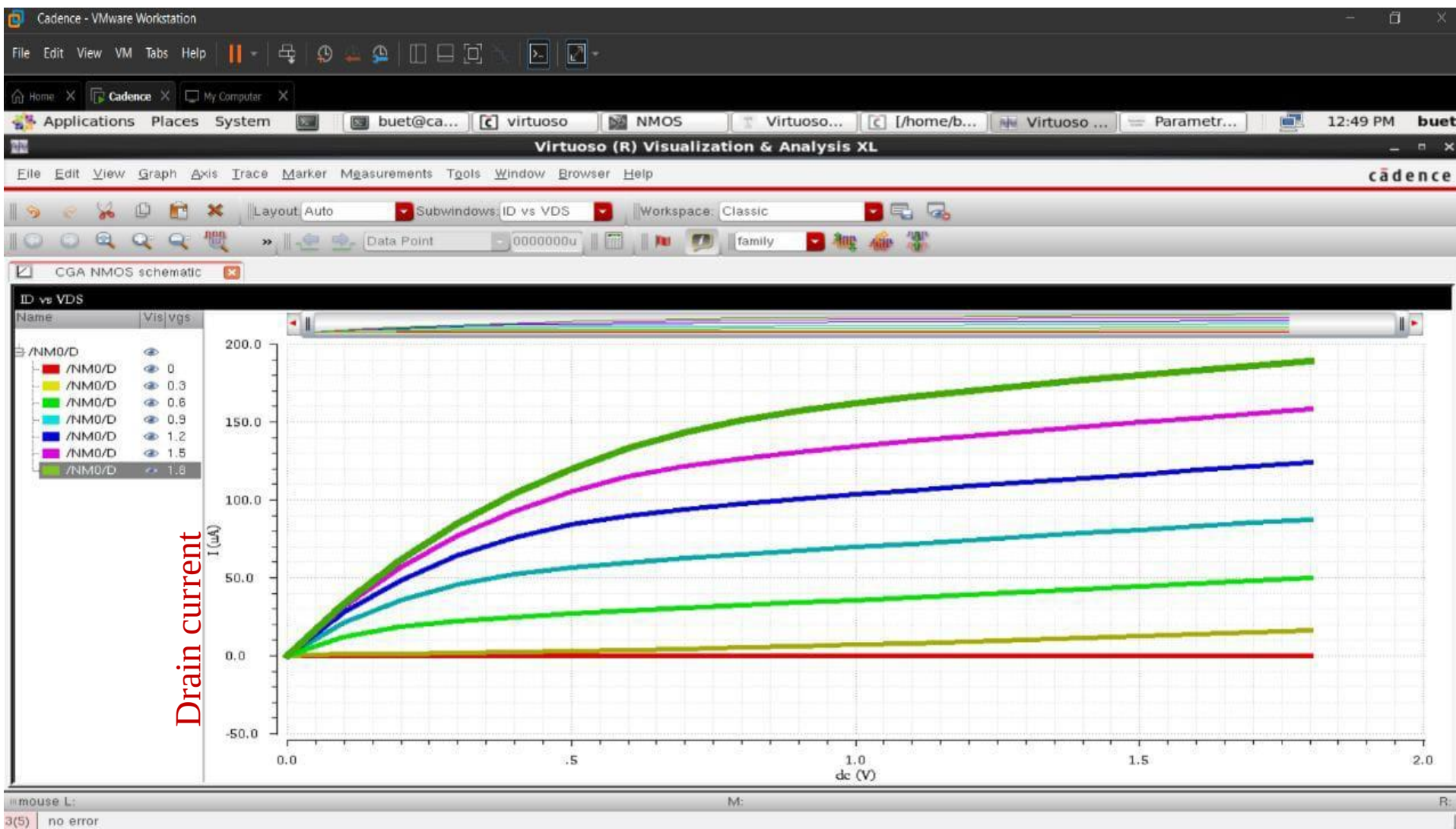


VGS

TRANSFER CHARACTERISTICS(ID vs VGS)

- Initially ID will be zero and further increases with VGS.
- With increase in length ID decreases.
- With increase in width ID increases.

DRAIN CHARACTERISTICS(I_D vs V_{DS})



V_{DS}

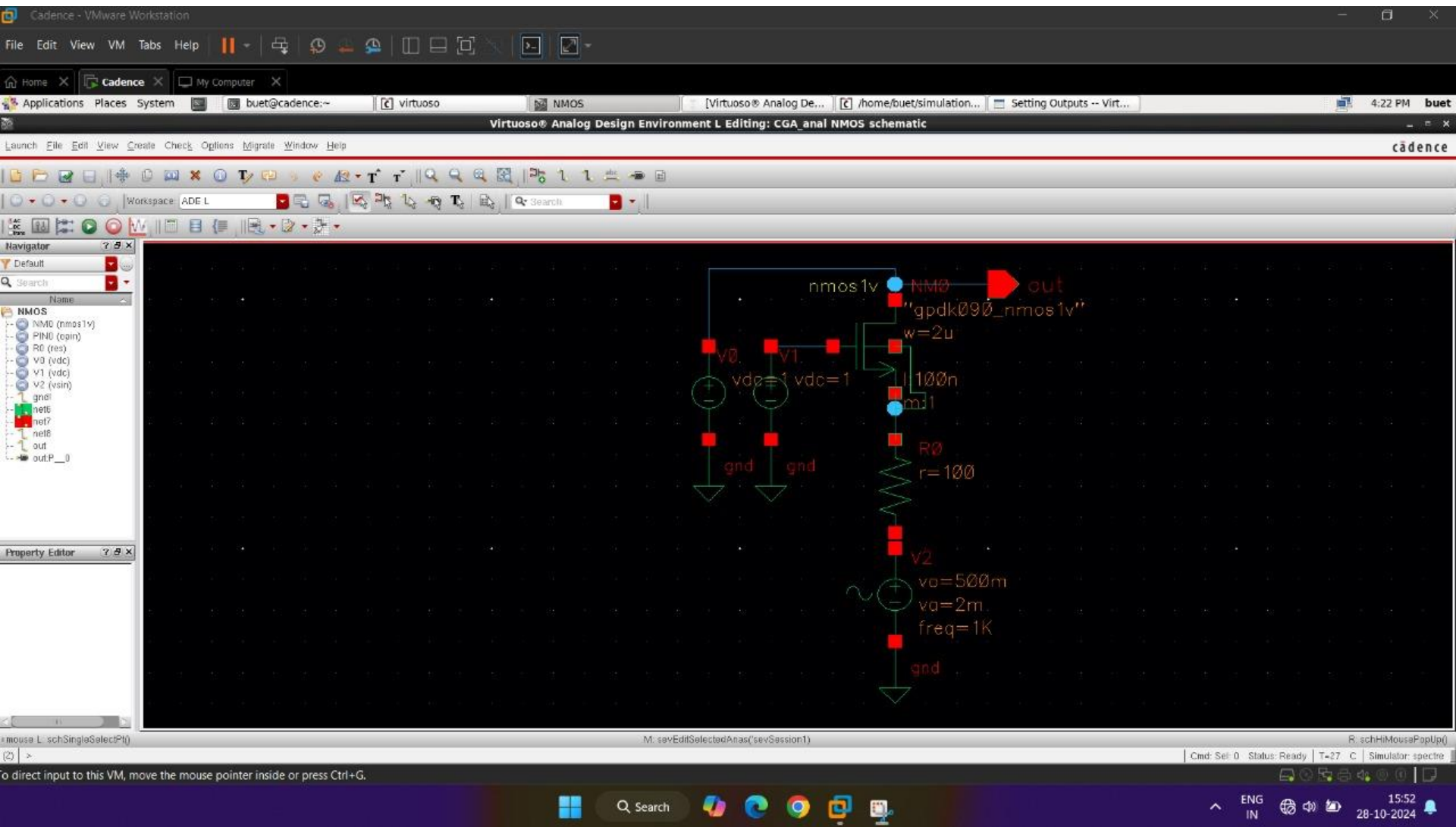
DRAIN CHARACTERISTICS(I_D vs V_{DS})

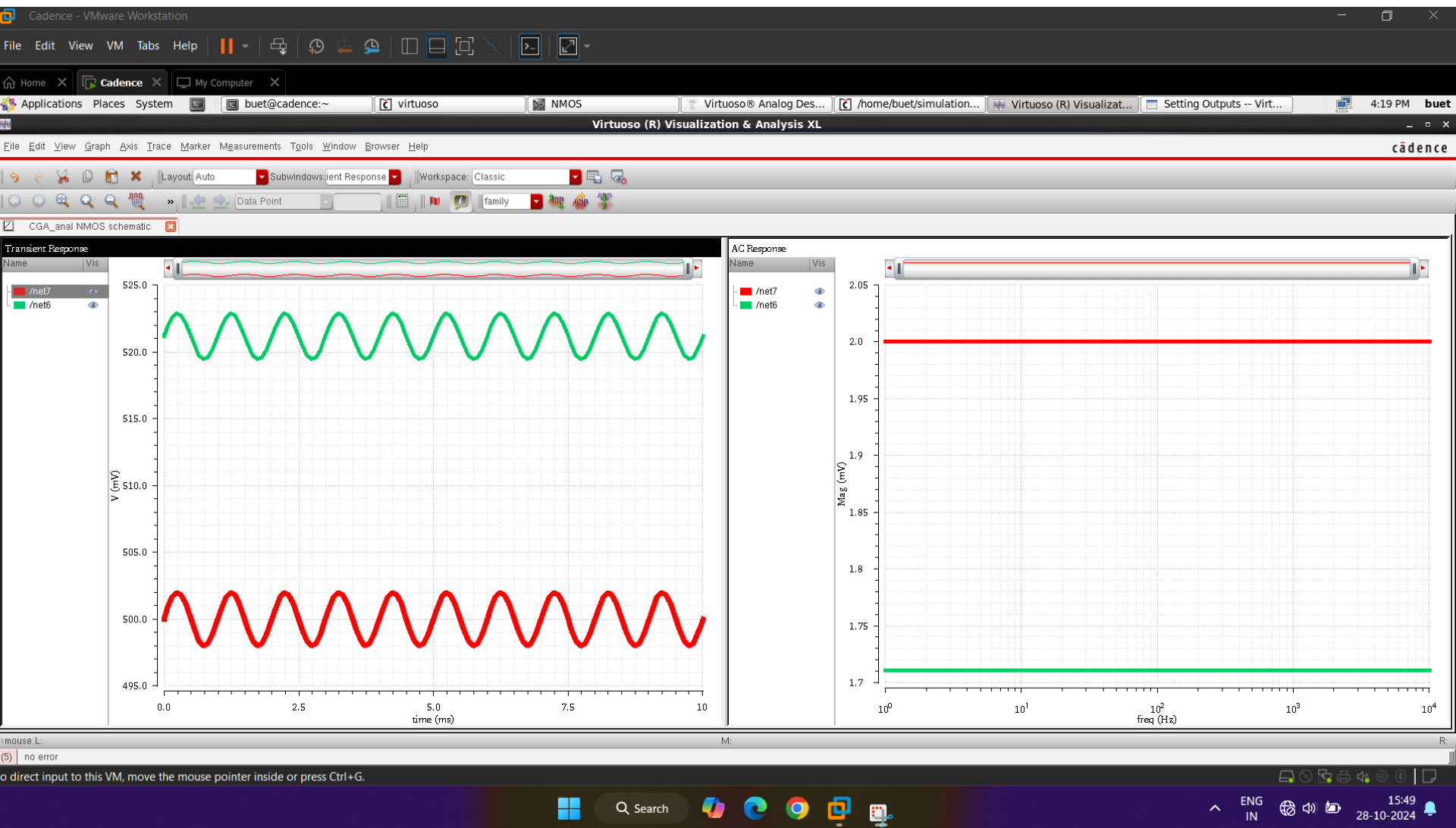
- I_D increases with V_{DS} .
- With increase in length I_D decreases.
- With increase in width I_D increases.

COMMON GATE AMPLIFIER

- A common-gate amplifier is an electronic amplifier configuration that uses a field-effect transistor (FET), where the gate terminal is common to both the input and output signals (typically held at a fixed voltage).
- The input signal is applied to the source terminal, and the output signal is taken from the drain terminal. This configuration is one of the three fundamental FET amplifier topologies.

COMMON GATE AMPLIFIER





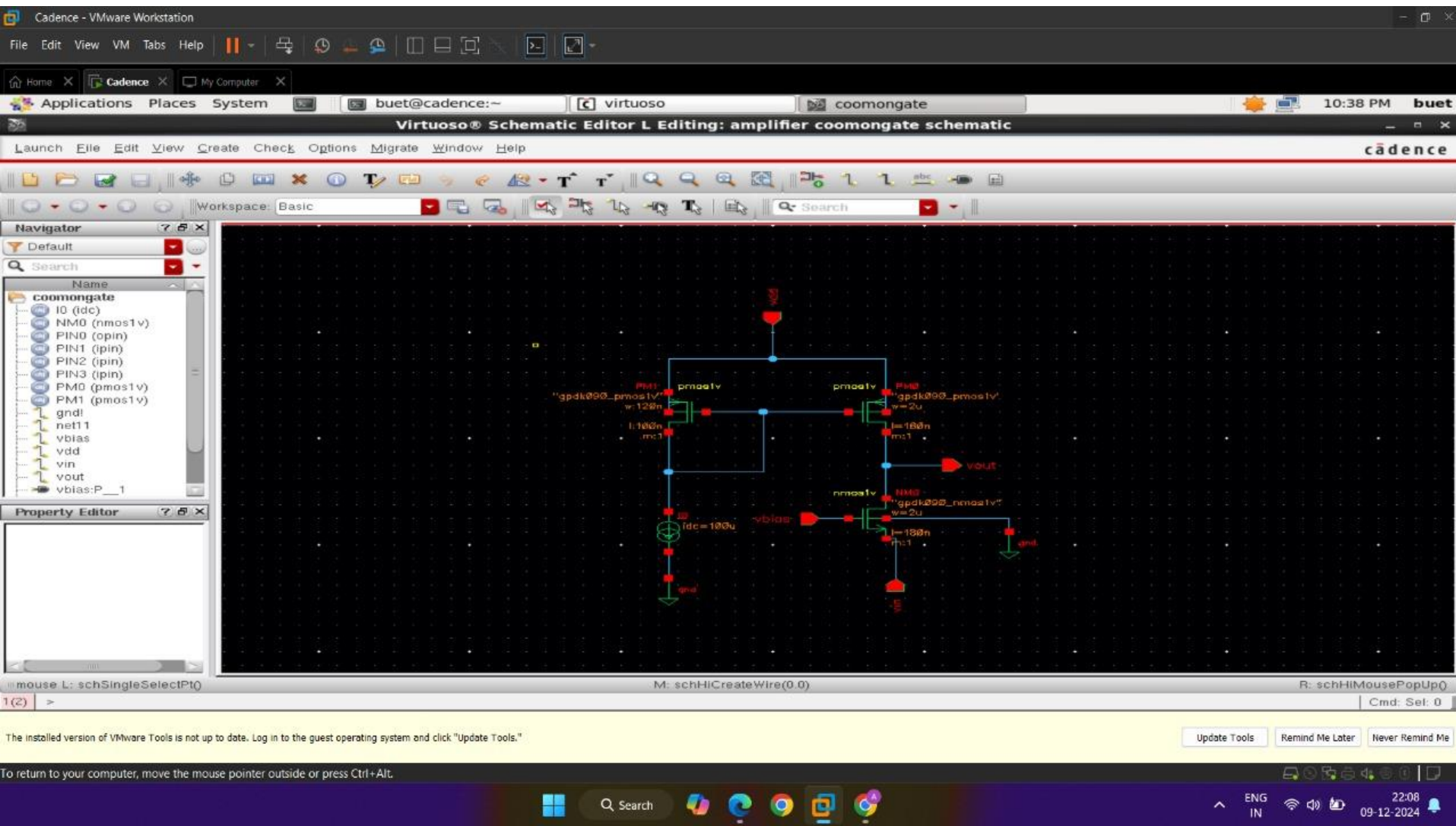
Voltage gain (Manual calculation)

$$\begin{aligned}V_{in} &= 502\text{mV} - 498.5\text{mV} \\ &= 3.5\text{mV}\end{aligned}$$

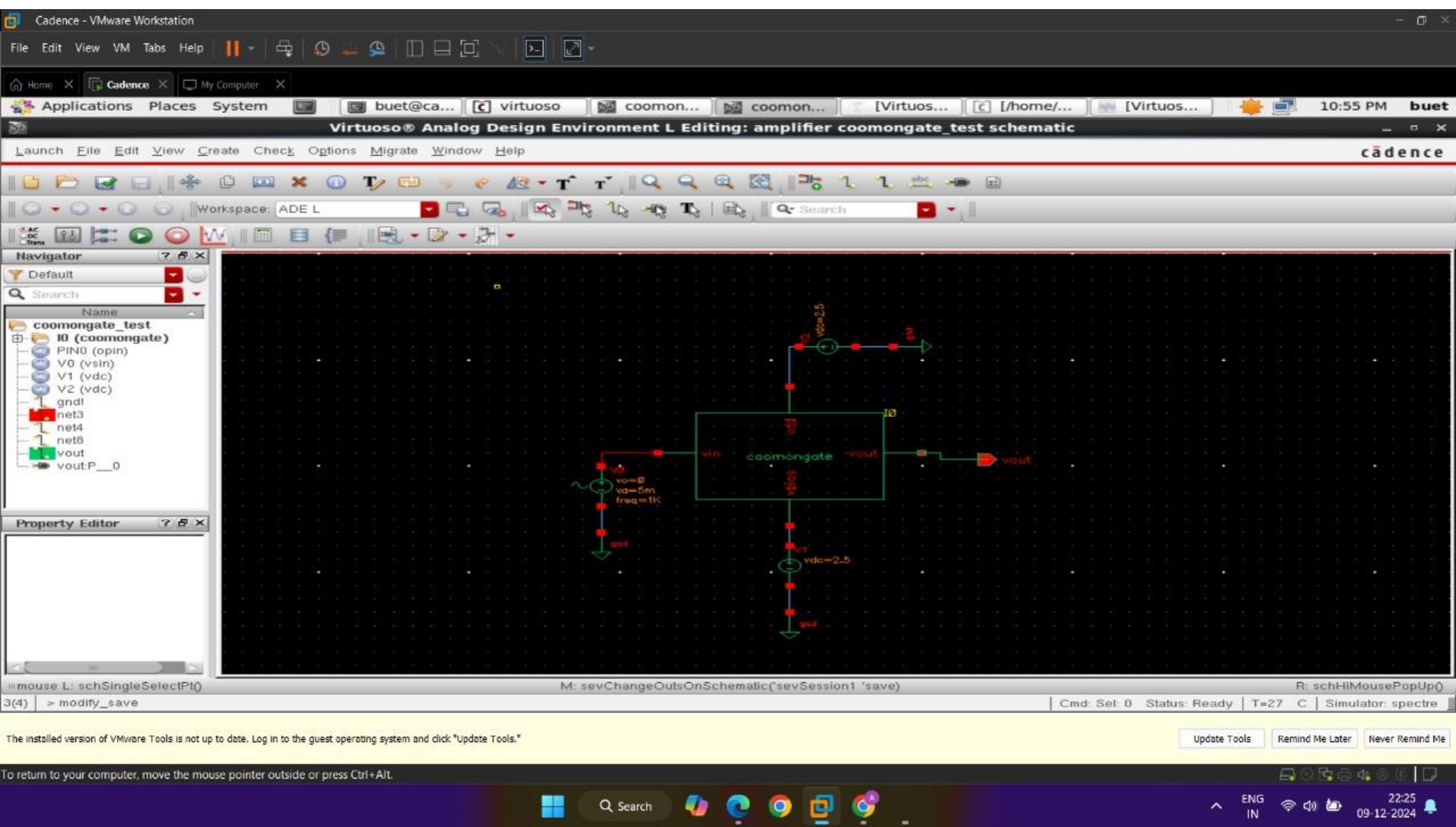
$$\begin{aligned}V_{out} &= 523\text{mV} - 519\text{mV} \\ &= 4\text{mV}\end{aligned}$$

$$\begin{aligned}\text{Gain} &= V_{out}/V_{in} \\ &= 1.148\end{aligned}$$

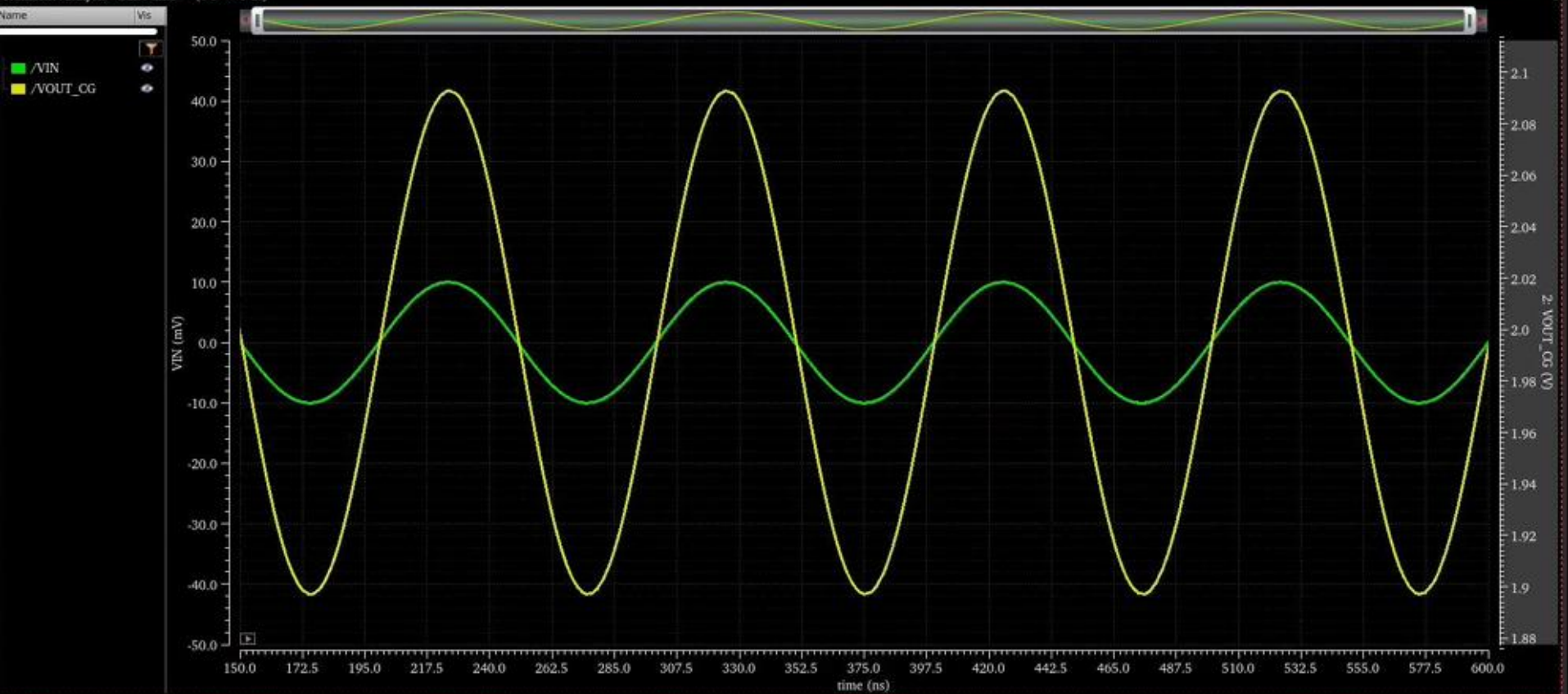
SCHEMATIC OF CGA



TESTBENCH OF CGA



Transient Analysis 'tran': time = (0 s -> 1 us)

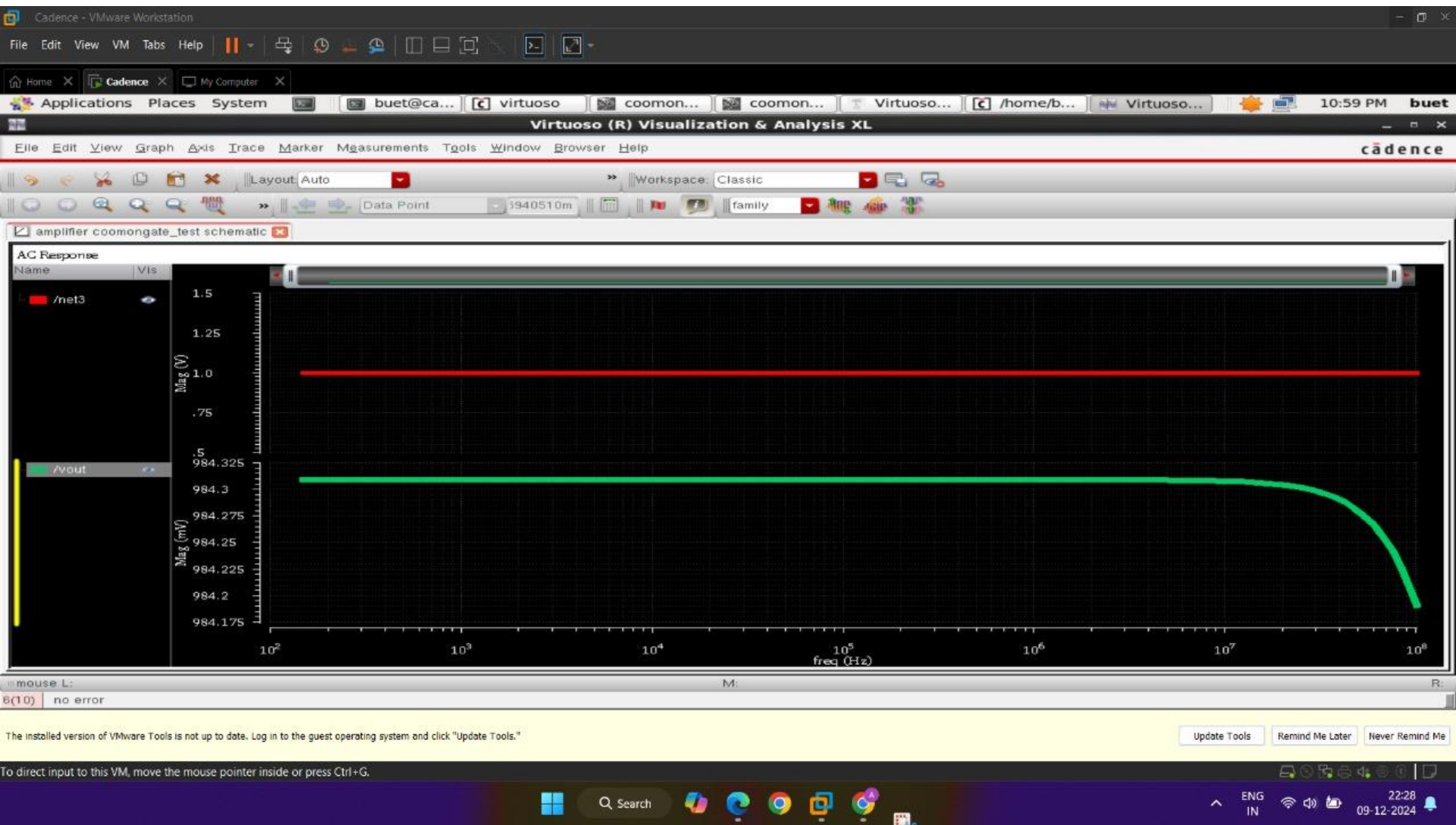


Voltage gain (Manual calculation)

$$\begin{aligned}V_{in} &= 10\text{mV} - (-10)\text{mV} \\ &= 20\text{mV}\end{aligned}$$

$$\begin{aligned}V_{out} &= 40\text{mV} - (-40)\text{mV} \\ &= 80\text{mV}\end{aligned}$$

$$\begin{aligned}\text{Gain} &= V_{out}/V_{in} \\ &= 4\end{aligned}$$



USING LTSPICE FOR A COMMON GATE AMPLIFIER

LTspice is a powerful, free circuit simulation tool that allows you to design, analyze, and test circuits virtually. For a common-gate amplifier, LTspice helps in visualizing performance through simulations like DC operating point analysis, AC analysis, and transient analysis.

Scenario	Preferred Tool	WHY?
Quick prototyping and validation.	LTspice	Easy setup, fast results, and intuitive interface.
Industry standard, process aware design.	Cadence	Foundry-specific models, high precision, and full PDK support.
Cross validation of results.	Both	Ensures consistency and correctness of simulations.

LTspice - [Draft1]

File Edit Hierarchy View Simulate Tools Window Help

.model nmos1 nmos (Vto=0.7V W=10um L=5um lambda=0.4 gamma=0 kp=10u)

.meas vin pp V(vin)

.meas vout pp V(vout)

.meas gain PARAM vout/vin

.op

.tran 0 1 0.8m 0.01m

VDD

V1

5

V2

1.2

vin

R1

920k

M1

nmos

R2

10

V3

SINE(0 1mV 5k)

C1

10μ

vout

LTspice - [Draft1]

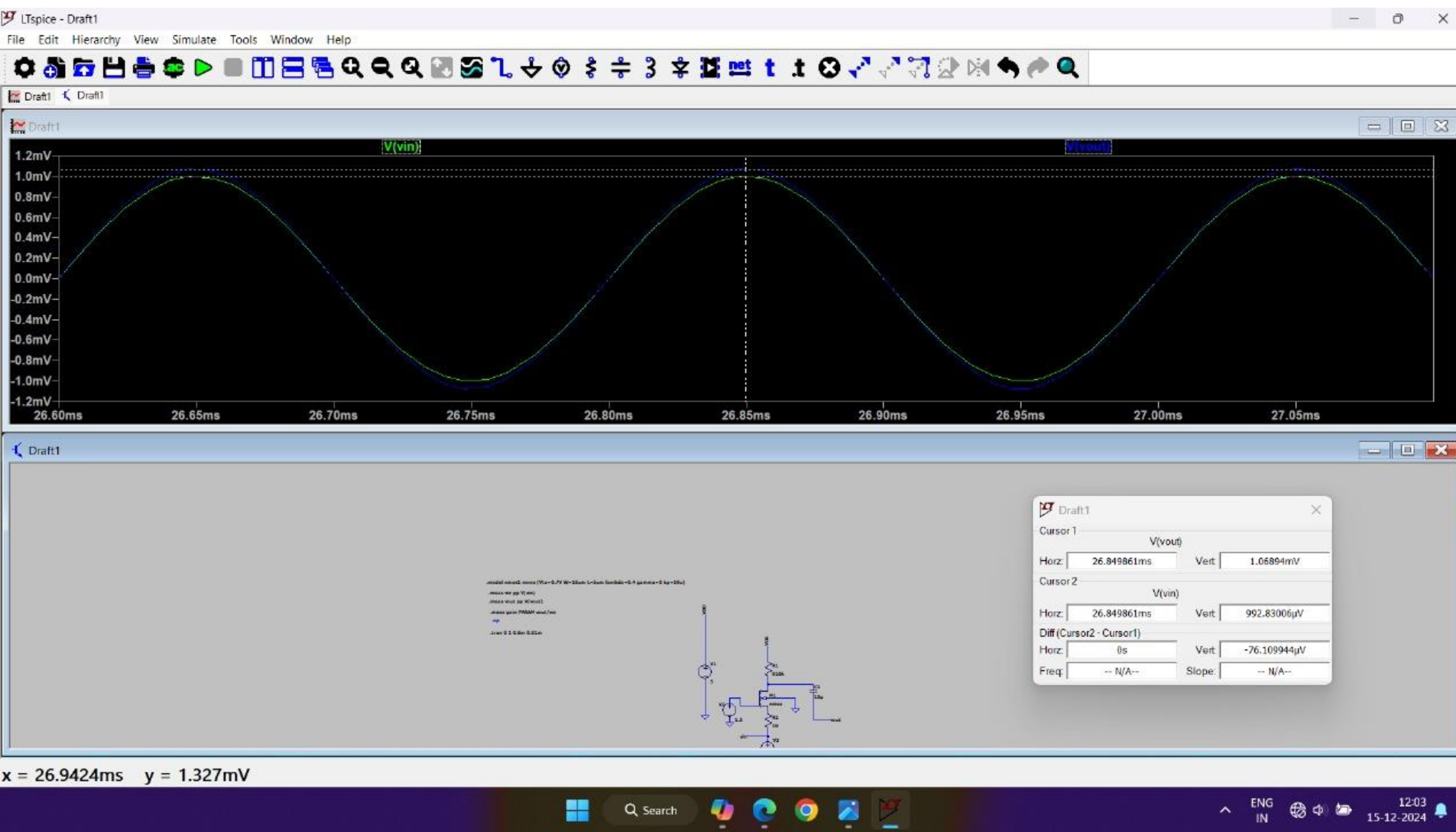
File Edit Hierarchy View Simulate Tools Window Help

.model nmos1 nmos (Vto=0.7V W=10um L=5um lambda=0.4 gamma=0 kp=10u)
 .meas vin pp V(vin)
 .meas vout pp V(vout)
 .meas gain PARAM vout/vin
 .op
 .tran 0 1 0.8m 0.01m

C:\Users\anush\OneDrive\Documents\LTspice\Draft1.asc
 --- Operating Point ---
 V(vdd): 5 voltage
 V(n001): 0.116278 voltage
 V(vout): 1.16277e-06 voltage
 V(n002): 1.2 voltage
 V(vin): 0 voltage
 V(n003): 5.30839e-05 voltage
 Id(M1): 5.30839e-06 device_current
 Ig(M1): 0 device_current
 Ib(M1): -1.2624e-13 device_current
 Is(M1): -5.30839e-06 device_current
 I(C1): 1.16277e-18 device_current
 I(R1): 5.30839e-06 device_current
 I(R2): 5.30839e-06 device_current
 I(V1): -5.30839e-06 device_current
 I(V2): 0 device_current
 I(V3): 5.30839e-06 device_current

Ready

Windows Taskbar: Search, File Explorer, Edge, Chrome, VS Code, LTspice
 System Tray: ENG IN, 12:04, 15-12-2024



Voltage gain (Manual calculation)

$$\begin{aligned}V_{in} &= 1\text{mV} - (-1)\text{mV} \\ &= 2\text{mV}\end{aligned}$$

$$\begin{aligned}V_{out} &= 1.5\text{mV} - (-1.5)\text{mV} \\ &= 3\text{mV}\end{aligned}$$

$$\begin{aligned}\text{Gain} &= V_{out}/V_{in} \\ &= 1.5\end{aligned}$$

SUMMARY OF RESULTS

Parameter	Transient analysis	AC analysis
Voltage gain	Amplifies input signal.	Constant in mid-band, reduce at high frequencies.
Phase	Output is in phase with input.	Phase shifts occur at high frequencies.
Rise/Fall time	Determined by circuit bandwidth.	Faster with reduced parasitic.
Bandwidth	Not directly visible, but impacts transition speeds.	Defined by the frequency response curve.
Stability	Stable response to transient inputs.	Stability depends on impedance and feedback.

CONCLUSION

- Technically, the common-gate amplifier is an essential building block in high-frequency amplification and signal processing.
- Its low input impedance, high output impedance, and moderate to high voltage gain make it invaluable in applications requiring impedance matching and current amplification.
- Its usage in low-noise amplifiers and RF systems underscores its importance in ensuring high-performance systems.

REFERENCES

1. Tae Wook Kim,“A Common gate amplifier with Transconductance Nonlinearity Cancellation and Its High-Frequency Analysis Using the Volterra Series” Publish year-2009.
2. Ahmad Qassem, Mohamed El-Nozahi, Hani Ragai,“Analysis and design of wideband common-gate low noise amplifier” Publish year-2015.
3. Piotr Legutko,“ The Analysis and Research of the Integrated, 30 A MOSFET Gate Driver Dedicated to High-Frequency Applications” Publish year-2024.

THANK YOU